

# STEEL PILES

| Type       | Section | Minimum number of piles used | Approx. minimum diameter at internal face of wall (metres) | Approx. maximum driving length at minimum diameter (metres) |
|------------|---------|------------------------------|------------------------------------------------------------|-------------------------------------------------------------|
| Larssen    | 6W      | 40                           | 6.47                                                       | 6                                                           |
| Larssen    | 9W      | 40                           | 6.42                                                       | 9                                                           |
| Larssen    | 12W     | 40                           | 6.38                                                       | 9                                                           |
| Larssen    | 16W     | 40                           | 6.34                                                       | 15                                                          |
| Larssen    | 20W     | 40                           | 6.28                                                       | 15                                                          |
| Larssen    | 25W     | 40                           | 6.23                                                       | 15                                                          |
| Larssen    | 32W     | 40                           | 6.23                                                       | 15                                                          |
| Larssen    | 3       | 40                           | 4.85                                                       | 9                                                           |
| Larssen    | 4A      | 40                           | 4.71                                                       | 15                                                          |
| Larssen    | 6       | 40                           | 4.91                                                       | 15                                                          |
| Frodingham | 1N      | 120                          | 18.36                                                      | 6                                                           |
| Frodingham | 1BXN    | 120                          | 18.10                                                      | 6                                                           |
| Frodingham | 2N      | 120                          | 18.32                                                      | 9                                                           |
| Frodingham | 3N, 3NA | 120                          | 18.30                                                      | 15                                                          |
| Frodingham | 4N      | 120                          | 18.28                                                      | 15                                                          |
| Frodingham | 5       | 120                          | 16.07                                                      | 15                                                          |

## EFFECTIVE LIFE

The effective life of unpainted or otherwise unprotected steel piling to BS 4360 Grades 43A and 50A depends upon the combined effects of imposed stresses and corrosion.

Performance is clearly optimised with combinations of high stress and low corrosion rate. Although other stress and corrosion combinations are less favourable, good design can ensure acceptable lives. The use of grade 50A steel at grade 43A stresses can also increase effective life. Copper-bearing steels are marginally advantageous but only in atmospheric exposures.

The table below indicates the minimum effective life for various environments assuming that maximum permissible design stress occurs in the zone under consideration. For regions of lower stress, the permissible loss in thickness may be increased and appropriate effective lives calculated from the corrosion data for each service environment.

### Minimum Effective Life for Maximum Stress in Various Service Environments

| Section type |            | Environments                                                         |                          |                            |                                    |             |
|--------------|------------|----------------------------------------------------------------------|--------------------------|----------------------------|------------------------------------|-------------|
| Frodingham   | Larssen    | Permissible loss in thickness (mm) at position of max stress in pile | Atmospheric corrosion    | Splash and low water zones | Sea water immersion and tidal zone | Underground |
|              |            |                                                                      | Corrosion rates, mm/year |                            |                                    |             |
|              |            |                                                                      | 0.05 (mean)              | 0.09 (mean)                | 0.05 (mean)                        | 0.03 (max)  |
|              |            |                                                                      | Effective life in years  |                            |                                    |             |
|              | 6W, 9W     | 3.5                                                                  | 70                       | *                          | *                                  | 117         |
|              | 12W, 16W   | 4.5                                                                  | 90                       | 50                         | 90                                 | 120 +       |
|              | 20W, 3, 4A | 5.0                                                                  | 100                      | 55                         | 100                                | 120 +       |
| 1N           |            | 5.2                                                                  | 104                      | 58                         | 104                                | 120 +       |
| 2N           |            | 5.4                                                                  | 108                      | 60                         | 108                                | 120 +       |
| 3NA          |            | 5.5                                                                  | 110                      | 60                         | 110                                | 120 +       |
| 3N           |            | 5.9                                                                  | 118                      | 65                         | 118                                | 120 +       |
|              | 25W, 32W   | 6.0                                                                  | 120                      | 67                         | 120                                | 120 +       |
|              | All No. 6  | 7.0                                                                  | 120 +                    | 78                         | 120 +                              | 120 +       |
| 1BXN         |            | 7.2                                                                  | 120 +                    | 80                         | 120 +                              | 120 +       |
| 4N           |            | 7.4                                                                  | 120 +                    | 82                         | 120 +                              | 120 +       |
| 5            |            | 8.9                                                                  | 120 +                    | 99                         | 120 +                              | 120 +       |

\* Not suitable for Marine Environment.